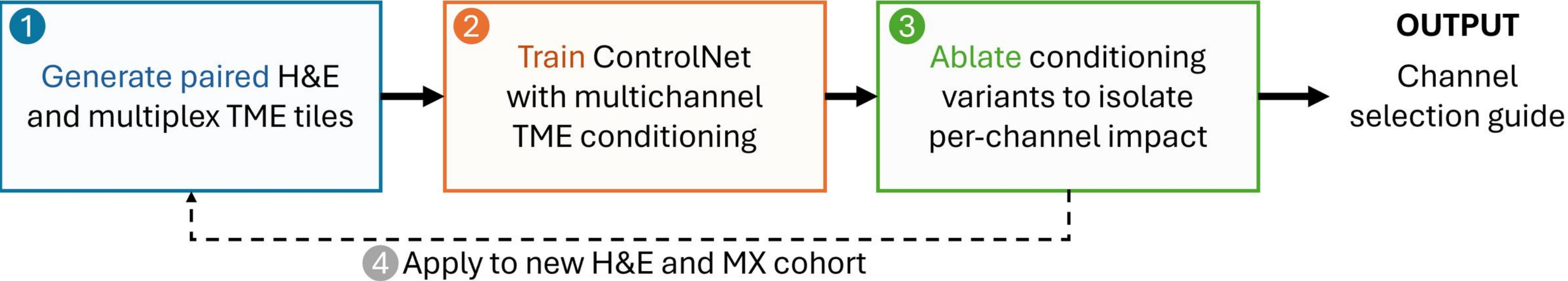


Main figures - paper preview (fig1-fig4, filled to common column width, 150 dpi)

(overview_workflow.png)



Graphical abstract. A three-stage pipeline turns paired H&E + multiplex tiles into a per-channel conditioning guide. (1) Generate paired H&E and multiplex TME tiles; (2) train a ControlNet with multichannel TME conditioning; (3) ablate conditioning variants to isolate per-channel impact, yielding the **output** : a channel-selection guide. The dashed return path (4) applies the trained model to a new, unpaired H&E + MX cohort. Stage colors (blue / orange / green) act as a paper-wide section code matching fig1-fig4.

(fig1_approach_data.png)

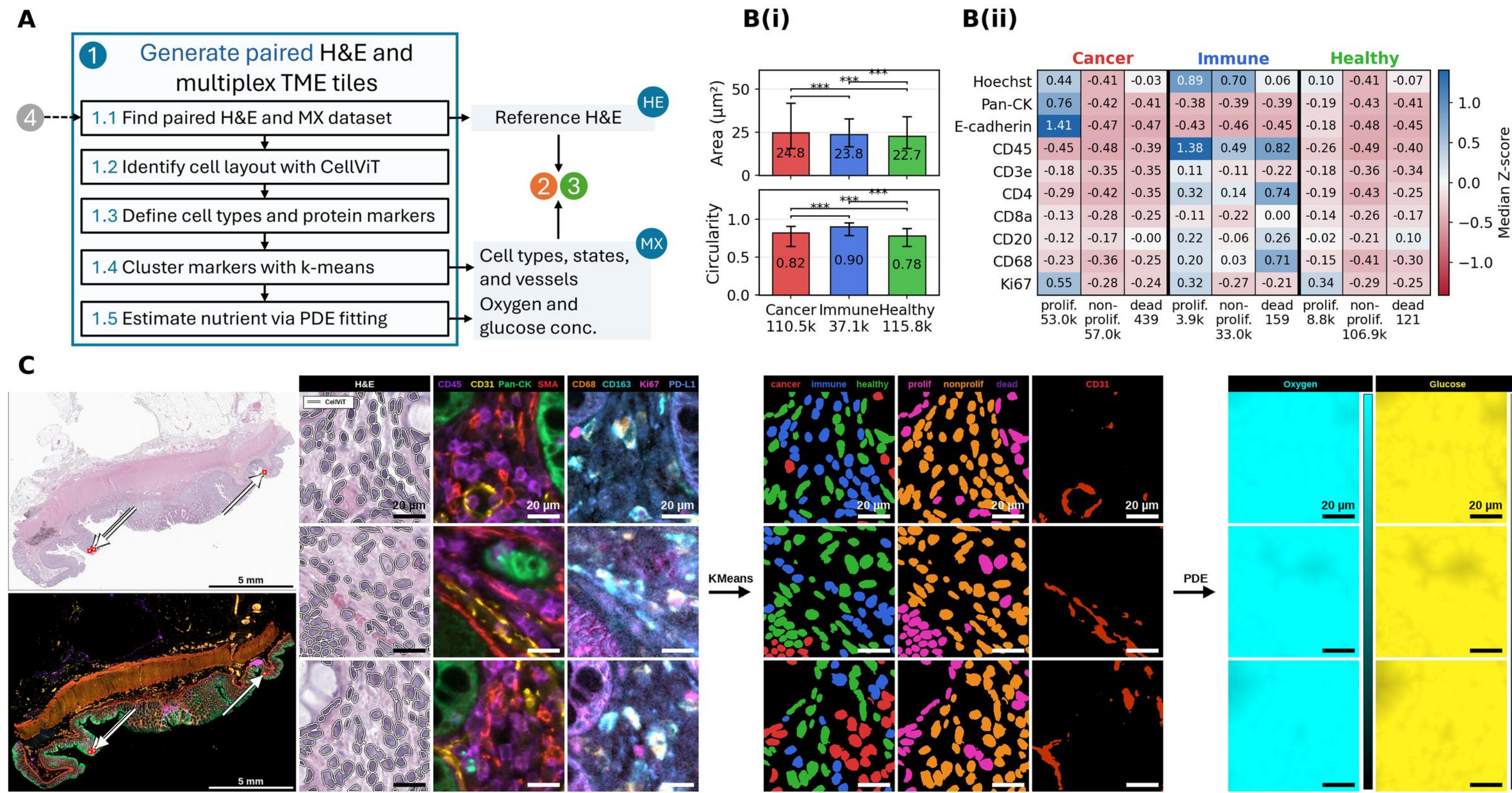


Figure 1. Generating paired H&E + multiplex TME tiles and validating the derived cell, state, and nutrient maps (Stage 1). (A) Stage-1 pipeline: paired H&E / multiplex (MX) tiles are matched, cells are segmented with CellViT, cell types and protein markers are defined, markers are clustered with k-means into cell types and states, and nutrient (oxygen, glucose) fields are estimated by PDE fitting. (B) Validation of the derived labels. (i) Per-cell-type morphology — nuclear area (μm^2) and circularity — for the cancer, immune, and healthy populations (mean $\pm$ SD; per-group n shown; pairwise significance marked). (ii) Mean marker z-scores per assigned cell type (columns: cancer / immune / healthy; rows: Hoechst, Pan-CK, E-cadherin, CD45, CD4, CD8, CD3, CD68, Ki67), color-scaled from red (low) to blue (high); the block-diagonal structure confirms lineage-consistent assignments. (C) Real-data montage for one specimen: whole-slide image $\rightarrow$ registered H&E with CODEX immunofluorescence channels $\rightarrow$ k-means cell segmentation/typing $\rightarrow$ PDE-estimated oxygen and glucose fields.

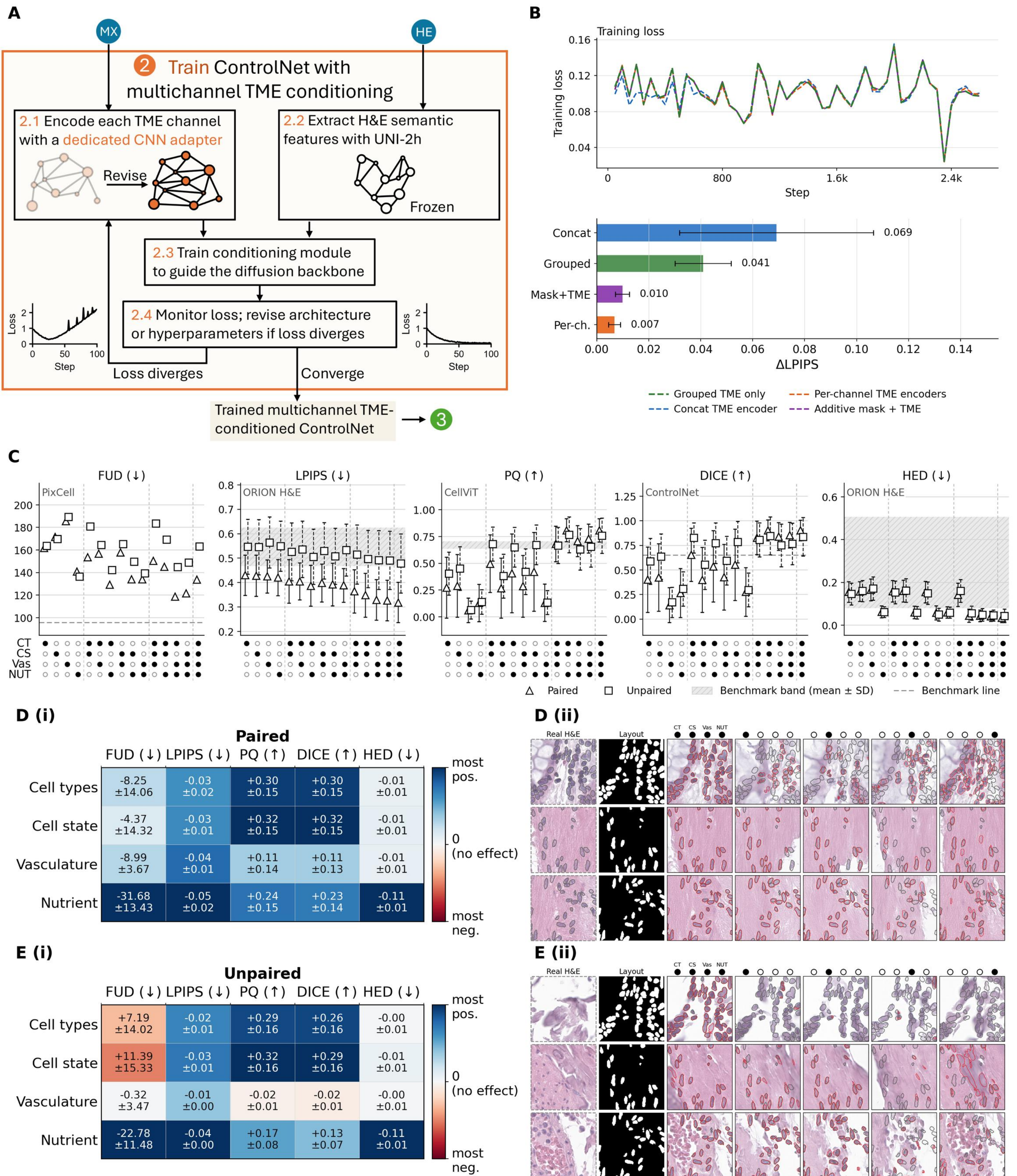


Figure 2. Training the multichannel TME-conditioned ControlNet and characterizing its performance (Stage 2). (A) Stage-2 schematic: each TME channel is encoded by a dedicated CNN adapter, H&E semantic features come from a frozen UNI-2h, and a conditioning module guides the frozen diffusion backbone; architecture and hyperparameters are revised when the loss diverges. (B) Training loss versus optimization step (top) and Δ LPIPS of the conditioning-architecture variants relative to baseline (bottom; Concat, Grouped, Mask + TME, Per-channel), with the shared variant legend. (C) Per-tile performance trade-offs across the 32 channel-group on/off conditions for five metrics — FUD ($\downarrow$), LPIPS ($\downarrow$), PQ ($\uparrow$), DICE ($\uparrow$), HED ($\downarrow$); triangles = paired, squares = unpaired inference; shaded band = benchmark mean $\pm$ SD. (D) Paired-inference channel effects: per-group effect-size heatmap (mean $\pm$ SD change attributable to including Cell types, Cell state, Vasculature, or Nutrient channels; blue = positive, red = negative) shown beside qualitative leave-one-out tiles for the same split. (E) Unpaired-inference channel effects, shown with the same heatmap scale and qualitative leave-one-out layout as in D.

(fig3_uni_decomposition_v2.png)

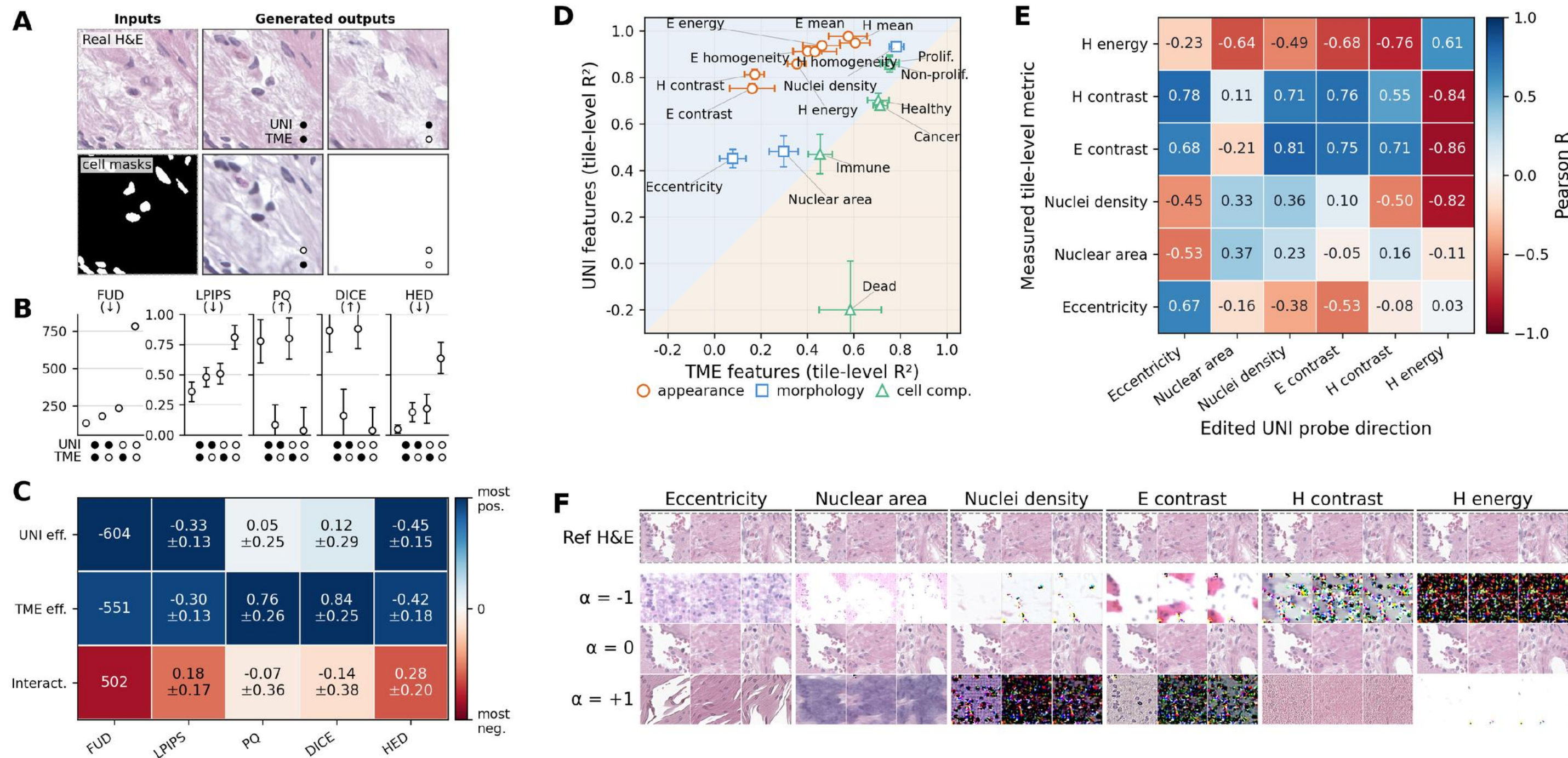
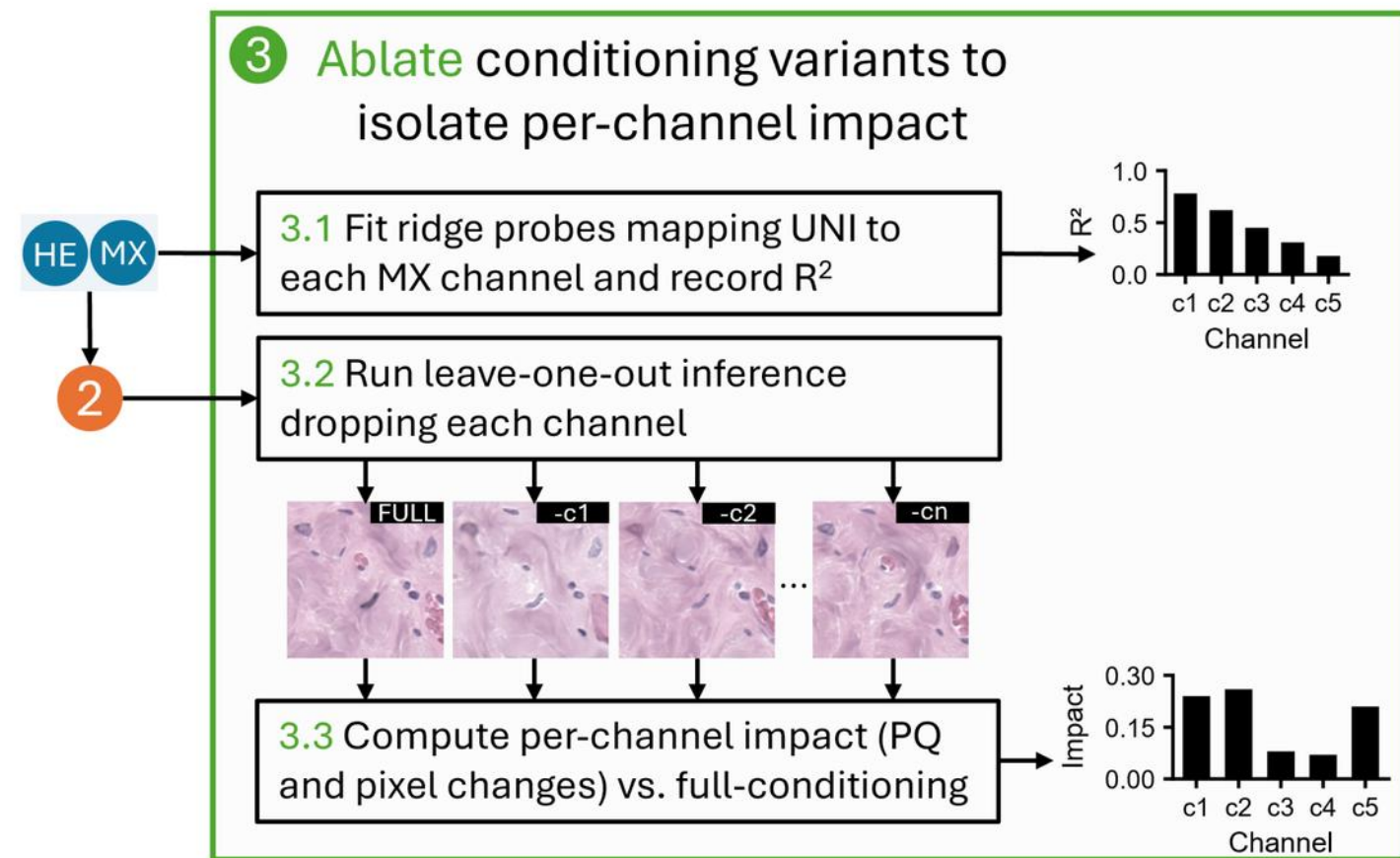


Figure 3. UNI conditioning and TME conditioning contribute near-orthogonally, and frozen UNI embeddings linearly encode interpretable H&E morphology. (A) A representative tile: model inputs (real H&E reference, cell masks) and generated outputs under the four UNI/TME on-off settings (the dot pair beside each row marks which of UNI, TME is active). **(B)** Per-tile mean $\pm$ SD for the four conditions across five metrics (FUD/LPIPS/PQ/DICE/HED); the 2-dot label encodes UNI/TME state. **(C)** Two-way decomposition of each metric: UNI effect (full – TME-only), TME effect (full – UNI-only), and UNI $\times$ TME interaction, color-scaled blue (most positive) to red (most negative) with values printed per cell. **(D)** Tile-level decodability of each H&E attribute from frozen UNI features (y-axis) versus from the spatial TME channels (x-axis), held-out R²; points colored and shaped by family (appearance, morphology, cell composition); the shaded diagonal marks attributes equally recoverable from either source. **(E)** Specificity matrix for UNI probe edits: each column edits the embedding along one attribute's probe direction, each row reports the Pearson correlation with the measured change in a tile-level morphology metric (–1 red to +1 blue); a strong diagonal indicates isolated control. **(F)** Targeted edit sweeps for six attributes (Eccentricity, Nuclear area, Nuclei density, E contrast, H contrast, H energy): the top row is the real reference H&E, the lower rows are H&E generated after shifting the UNI embedding along that probe direction by $\alpha = -1, 0, +1$.

(fig4_per_channel_impact.png)

A



B

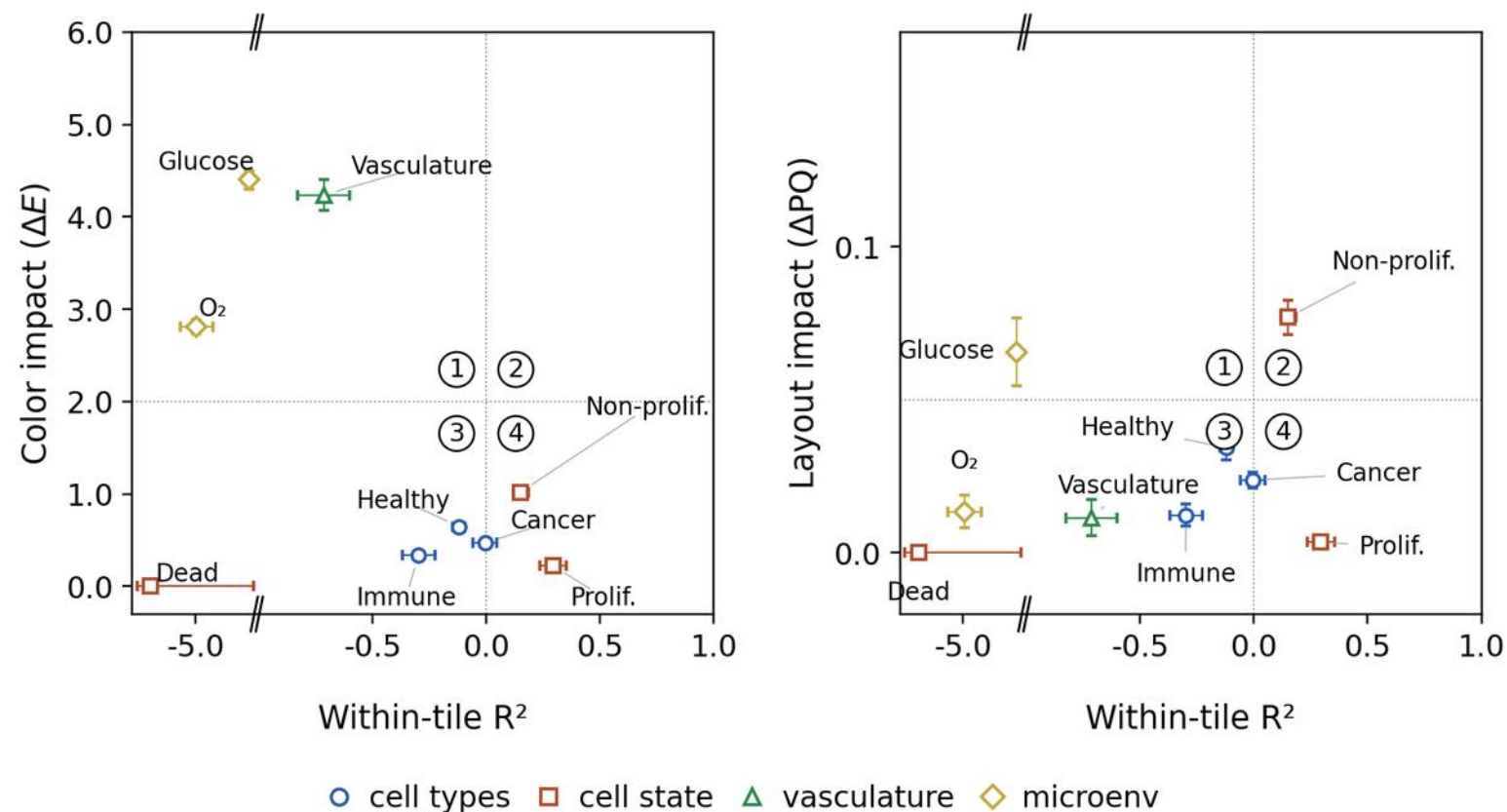


Figure 4. Relating H&E decodability to generative impact yields a channel-selection guide (Stage 3). **(A)** Stage-3 schematic: ridge probes map UNI to each MX channel (recording R^2), leave-one-out inference drops each channel, and per-channel impact (PQ and pixel changes) is computed against full conditioning. **(B)** Decodability versus generative impact: color impact ΔE (left) and layout impact ΔPQ (right) plotted against within-tile decoding R^2 (UNI-2h spatial probe); points colored and shaped by group (cell types, cell state, vasculature, microenv), with dotted per-panel quadrant guides. The supporting decodability measurements and the ranked per-channel impacts are provided in the SI (Fig. S4).